VISION-BASED PRECISION POSE ESTIMATION FOR AUTONOMOUS FORMATION FLYING

UE24CS352A • Machine Learning Mini-Project Report

Team: Manish Bala and Manasvi D R | Computer Science & Engineering, PES University

1. Problem Statement and Motivation

This project explores a camera-based pipeline for detecting nearby drones, extracting image-space positions, estimating approximate depth from apparent drone width, and summarizing formation spacing. It is a software prototype: it does not control real drones or demonstrate flight-ready autonomous navigation.

2. Dataset and Setup

The detector was trained with the Roboflow export *Multiple Drones.v1i.yolov11*. Project records describe 640 × 640 RGB images and a test split of 96 frames with 277 labelled drone instances. Distance calculations assume a drone width W = 0.46 m and horizontal field of view (HFOV) of 60°. For a 640-pixel-wide image, this corresponds to an assumed focal length of approximately 554 pixels. These are assumptions, not independently measured camera calibration values.

3. Ten-Phase Methodology

Stages	Purpose
1–2 • Detection and centroids	YOLO11n predicts drone boxes; box centres provide image coordinates (cx, cy).
3–4 • Relations and evaluation	Compare image-space positions and evaluate detector predictions on the test split.
5–6 • Visualization and mapping	Create annotated visualizations and normalize coordinates for comparison across resolutions.
7–8 • Spacing and statistics	Compute pairwise geometry and summarize nearest-neighbour spacing and formation status.
9–10 • Charts and approximate depth	Generate statistical visualizations and use a pinhole approximation for depth and tracking HUD output.

Depth approximation: $Z \approx (f \times W) / w$, where w is detected box width in pixels. Lateral and vertical offsets are estimated from the box centre relative to the image centre. Estimates depend on box quality, target dimensions, and camera calibration.

4. Detector Metrics and Verification

Metric	Saved result	Interpretation
Precision	99.63%	Reported in saved project results; not independently reproduced here.
Recall	99.51%	Reported in saved project results; not independently reproduced here.
mAP@0.50	99.36%	Reported in saved project results.
mAP@0.50:0.95	85.12%	Reported in saved project results.

Local checks: verify_pipeline.py reported all 10 phases passed (6.08 seconds). The saved demo video was played and tracking appeared to work. These checks do not independently prove a clean rebuild from scratch.

RESULTS, LIMITATIONS & TEAM CONTRIBUTIONS

Vision-Based Precision Pose Estimation for Autonomous Formation Flying

5. Formation and Distance-Estimation Results

The saved Phase 10 output reports 96 frames and 277 detections. With the assumed 0.46 m drone width and 60° HFOV, detected box width had median 30.1 px (range 8.4–154.6 px); estimated distance had median 8.5 m (range 1.6–30.2 m). These are formula-based estimates, not independent range measurements.

Formation status	Frames	Share of 96
Maintained	44	45.8%
Outside threshold / too far	34	35.4%
Single drone detected	12	12.5%
Too close	6	6.3%

Saved analysis reports median nearest-neighbour spacing of 5.9 drone-widths (IQR 4.1–9.4) and median estimated 3D nearest-neighbour spacing of 4.44 m.

6. Distance-Error Analysis: Interpretation

For 272 matched predicted/ground-truth boxes (mean IoU 0.915), the saved analysis reports MAE = 0.252 m, RMSE = 0.360 m, mean absolute percentage error = 2.5%, and 90th-percentile absolute percentage error = 5.2%. These compare distance estimates calculated from detected box widths with estimates calculated from ground-truth box widths using the same formula. **They measure sensitivity to bounding-box error, not accuracy against physically measured distances.** Independent validation requires known target width, measured distances, and camera calibration.

Key limitation: Real-world distance accuracy has not yet been independently validated. Do not present the 0.252 m figure as real-world ranging accuracy.

7. Edge-Case Stress Test

The local edge-case script was run on drone_original.jpg and produced eight variations. Original image: detected at confidence 0.48. Blur kernels 15 and 35: no detection. Brightness-scaled variants at 50% and 20%: detected at 0.31 and 0.25. Cropped-left, cropped-right, and cropped-top variants: detected at 0.34, 0.25, and 0.40. This small synthetic test does not establish general robustness in real flight conditions.

8. Limitations and Future Work

- Validate estimated depth against independently measured physical distances and calibrate the actual camera field of view.
- Investigate motion-blur failures and false positives on out-of-distribution scenes; broaden training data and include negative samples.
- Test a complete pipeline rerun from a clean environment with dependencies and source inputs available.
- Consider temporal filtering (e.g., an Extended Kalman Filter) to smooth frame-to-frame estimates before any real deployment.

9. Individual Contributions

Member	Contribution summary
Manish Bala	YOLO11n training and validation; 2D detection/centroid extraction; relative image-space bearing; coordinate normalization and earlier geometry pipeline; core repository work.
Manasvi D R	Monocular depth estimation; dimensionless nearest-neighbour spacing and formation-status logic; top-down radar projection; bounding-box error-sensitivity analysis; edge-case and calibration tooling.

Repository: github.com/manishribil/Vision-Based-Pose-Estimation
Reproducibility note: Results above are drawn from available saved project outputs and local checks. Clean, independent end-to-end reproduction and physical range validation remain outstanding.